Document Purpose

This document explains the function of the *temperature* and *Top_P* parameters used in the Bedrock Chat Application. These parameters were implemented in the Streamlit interface to give users control over the model's creativity and response consistency during text generation.

Temperature and Top_P Explanation

In generative AI models, **temperature** controls how random or creative the model's output is. A lower temperature (close to 0) makes the model more focused and deterministic — it repeats known and likely answers. A higher temperature (closer to 1) increases randomness and creativity, allowing more diverse or unexpected responses. In this project, adjusting the temperature slider in the Streamlit app changes how confidently or creatively Amazon Bedrock generates its responses about heavy machinery.

The **Top_P** parameter, also known as nucleus sampling, determines how much of the probability distribution the model considers when generating text. Instead of choosing from all possible words, the model looks only at the most likely ones that together have a cumulative probability equal to Top_P. For example, a Top_P of 0.9 means the model focuses on the top 90% most likely words. Lower Top_P values make outputs more focused, while higher values make them broader and more diverse. Together, **temperature and Top_P provide fine-grained control over the balance between accuracy and creativity** in the model's responses.